

Attention U-Net & ResNet-50 Classification

Weeks 5–6 Progress — Model Development

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Segmentation — 3-way comparison

Model	Params	Dice	IoU
Baseline U-Net	31.04M	0.842	0.727
U-Net + ResNet-34	24.44M	0.225	0.127
Attention U-Net	31.39M	0.965	0.932

Attention gates focus the skip connections on tumour regions; built on the baseline backbone to isolate the attention effect.

Attention U-Net — test predictions

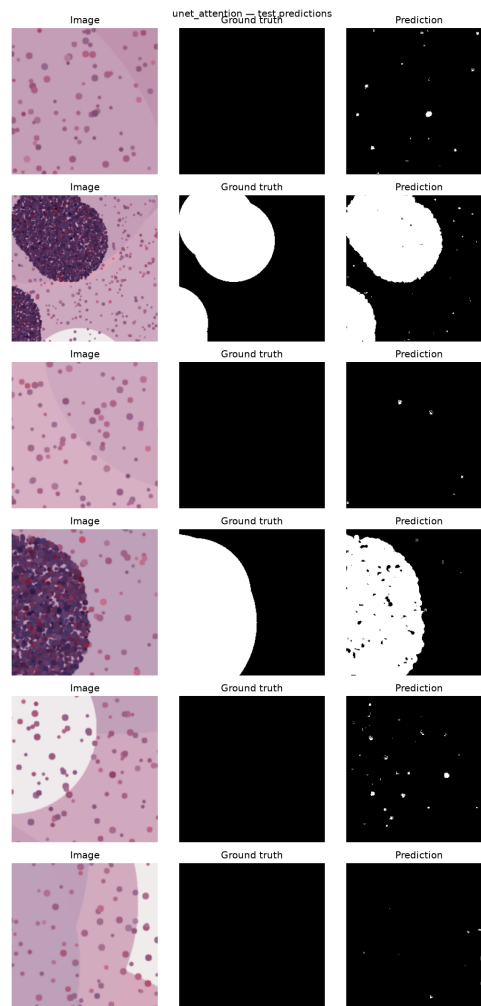


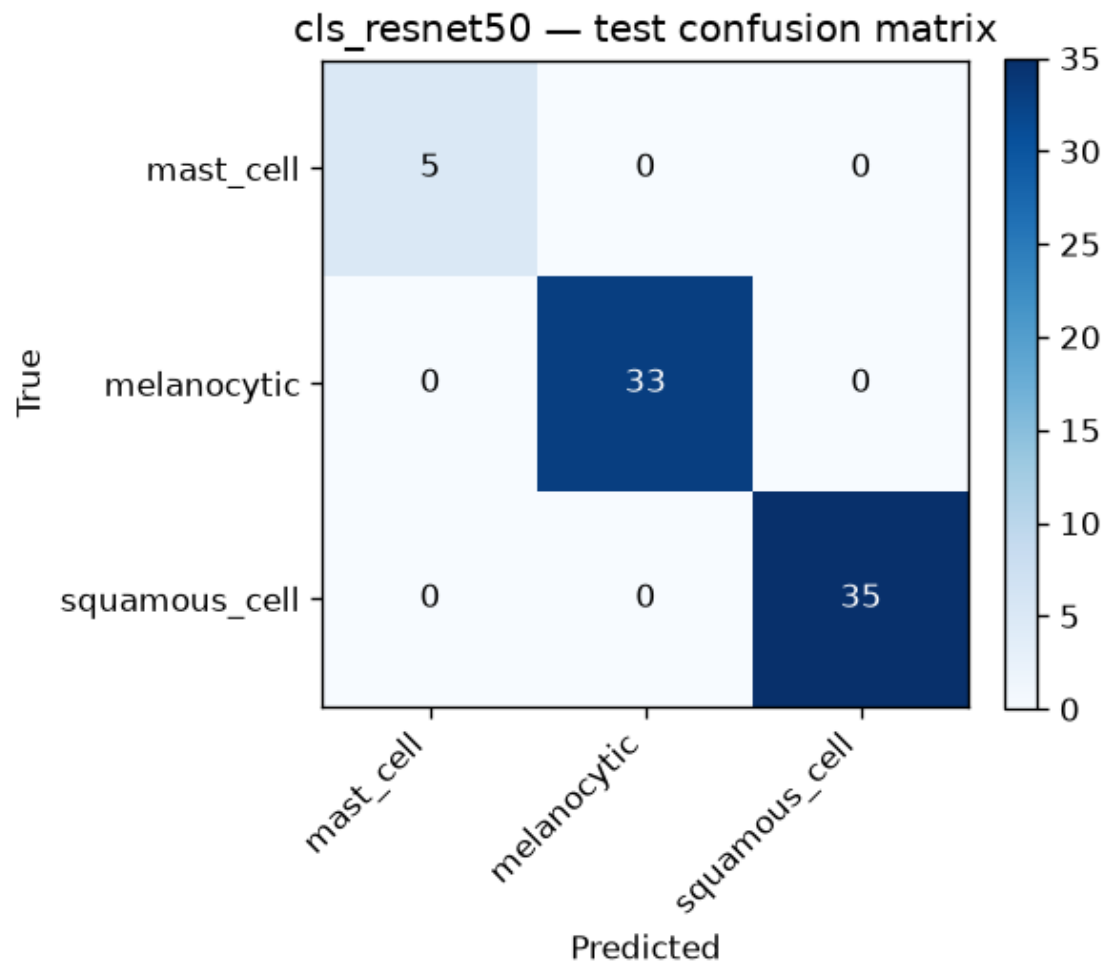
Image | ground-truth mask | predicted mask.

Classification — ResNet-50 (test)

Metric	Value
Accuracy	1.000
Macro F1	1.000
Macro precision	1.000
Macro recall	1.000
AUC-ROC	1.000

Classes: mast_cell, melanocytic, squamous_cell

ResNet-50 — confusion matrix



Diagonal = correct subtype predictions.

Next fortnight (Weeks 7–8)

- EfficientNet-B3 classifier + comparison with ResNet-50
- Run on real CATCH slides once downloaded
- Evaluation: 5-fold cross-validation, McNemar's test

Thank you

Questions & feedback welcome